

Curriculum Vitae

1) Personal information

Name: Yunan Hu, M.S.

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2) Education

<u>Degree</u>	<u>Institution</u>	<u>Date conferred</u>	<u>Major</u>
Ph. D.	University of North Carolina at Chapel Hill (UNC-CH), USA	Present	Microbiology and Immunology
M.S.	University of North Carolina at Chapel Hill (UNC-CH), USA	May 2023	Nutrition
B.S.	China Agricultural University (CAU), China	May 2021	Food Science and Engineering

3) Bibliography and products of scholarship

Peer-reviewed publications

Career Total = 10 Google Scholar h-index = 7 Google Scholar citations=280

1. **Y Hu**, A Rueppell, N Devpura, R Zasloff, J Arnold, J Roach, S Magness, J Bruno-Bárcena, MA Azcarate-Peril. Prebiotic β (1-4) Galacto-oligosaccharides Strengthen Intestinal Barrier Integrity via Mucus Production and Modulation of Tight Junction Genes. *Food Bioscience*. 2025;108052. DOI:10.1016/j.fbio.2025.108052
2. MR Aljumaah, J Roach, **Y Hu**, J Gunstad, MA Azcarate-Peril. Microbial dipeptidyl peptidases of the S9B family as host-microbe isozymes. *Sci Adv*. 2025 Apr 4;11(14):eads5721. doi: 10.1126/sciadv.ads5721.
3. **Y Hu**, MR Aljumaah, MA Azcarate-Peril. Galacto-Oligosaccharides and the Elderly Gut: Implications for Immune Restoration and Health. *Advances in Nutrition* 2024;15(8):100263. doi: <https://doi.org/10.1016/j.advnut.2024.100263>.
4. J Kang, R Li, **Y Hu**, X Huang, X-X Chen, B-Z Han. Microbial interactions in mixed-species biofilms on the surfaces of Baijiu brewing environments. *Food Research International* 2024;191:114698. doi: <https://doi.org/10.1016/j.foodres.2024.114698>.
5. S Tian, **Y Hu**, X Chen, C Liu, Y Xue, B Han. Green synthesis of silver nanoparticles using sodium alginate and tannic acid: characterization and anti-*S. aureus* activity. *International Journal of Biological Macromolecules* 2022;195:515-22. doi: <https://doi.org/10.1016/j.ijbiomac.2021.12.031>.
6. J Kang, **Y Hu**, L Jia, M Zhang, Z Zhang, X Huang, X Chen, B-Z Han. Response of microbial community assembly and succession pattern to abiotic factors during the second

round of light-flavor Baijiu fermentation. *Food Research International* 2022;162:111915. doi: <https://doi.org/10.1016/j.foodres.2022.111915>.

7. J Kang, **Y Hu**, Z Ding, L Ye, H Li, J Cheng, L Fan, H Zhao, B Han, X Zheng. Deciphering the Shifts in Microbial Community Diversity From Material Pretreatment to Saccharification Process of Fuyu-Flavor Baijiu. *Front Microbiol* 2021;12:705967. doi: 10.3389/fmicb.2021.705967.

8. **Y Hu**, X Huang, B Yang, X Zhang, Y Han, X-X Chen, B-Z Han. Contrasting the microbial community and metabolic profile of three types of light-flavor Daqu. *Food Bioscience* 2021;44:101395. doi: <https://doi.org/10.1016/j.fbio.2021.101395>.

9. 胡雨楠, 张钟晓, 陈小雪, 薛岩松, 韩北忠. 发酵时间对清香型白酒二萜基酒风味物质组成及含量的影响. *中国酿造* 2021;40(9):165-72.

10. X Chen, **Y Hu**, S Tian, B Han. Understanding the Interactions between *Staphylococcus aureus* and the Raw-Meat-Processing Environment Isolate *Klebsiella oxytoca* in Dual-Species Biofilms via Discovering an Altered Metabolic Profile. *Microorganisms* 2021;9.

Publications in process

1. **Y Hu**, J Arnold, et al. Mechanisms of Galectin 1 Induction by Galactooligosaccharides.

2. **Y Hu**, MR Aljumaah, J Roach, et al. Spatial ecology of the multi-kingdom human intestinal microbiome. Doi: <https://doi.org/10.21203/rs.3.rs-9927529/v1>

3. MR Aljumaah, I Isawi, J Roach, **Y Hu**, et al. Gut Microbial Dipeptidyl Peptidase-4 and Associated Prolyl Tripeptidyl Peptidases: A Novel Classification and Response to Inhibitors. Doi: <https://doi.org/10.21203/rs.3.rs-7312416/v1>

4. BR Wang, **Y Hu**, et al. Structural characterization and hepatoprotective mechanism of alkali-soluble *Poria cocos* polysaccharides modified by dynamic high-pressure microfluidization.

5. **Y Hu**, et al. *Lactobacillus* co-culture dynamics in response to different carbohydrates.

6. **Y Hu**, M Moore, et al. Synergistic Effects of β (1-4) Galactooligosaccharides Combined with Rationally Designed Consortia on Preventing *Clostridioides difficile* Infection.

Posters and conference proceedings

Career Total = 7

1. Conference: Gordon Research Conference (GRC): Lactic Acid Bacteria Biology, Symbioses and Applications, 2025, CA: (Oral presentation and poster) **Yunan Hu**, Madison Moore, Jeff Roach, Jeena James, Jason W. Arnold, Rita Tamayo, José M. Bruno-Bárcena, M. Andrea Azcarate-Perill. Pure prebiotics enable the persistence of probiotics and enhance their beneficial effects.

2. Conference: North Carolina Branch of the American Society for Microbiology (NC-ASM) 2024, NC: (Oral presentation) **Yunan Hu**: Galacto-oligosaccharides Enhance Intestinal Barrier Function.

3. *Conference: American Society for Nutrition (ASN) 2024, Boston:* (Poster) Jeena M James, **Yunan Hu**, Mashael Aljumaah, Esmat Rousta, Nivedita Pareek, Jeff Roach, Jason Arnold, Rita Tamayo, José Bruno-Bárcena, Andrea Azcarate-Peril. The Beneficial Effects of a Combination of Prebiotics and Bacterial Consortium on *Clostridioides difficile* Infection in a Mouse Model. 2024, Current Developments in Nutrition.
4. *Conference: American Society for Nutrition (ASN) 2023, Boston:* (Poster) **Yunan Hu**, Jason Arnold, Johanna Smeekens, Michael Kulis, Ellyce San Pedro, Scott Magness, Andrea Azcarate-Peril. Immunomodulatory Effects of Galactooligosaccharides. 2023, Current Developments in Nutrition.
5. *Conference: International Scientific Association for Probiotics and Prebiotics (ISAPP) 2023, Denver:* (Poster) **Yunan Hu**, Jason Arnold, Johanna Smeekens, Michael Kulis, Ellyce San Pedro, Scott Magness, Andrea Azcarate-Peril. Immunomodulatory Effect of Galacto-oligosaccharide on Lgals-1 Gene Expression.
6. *NC Microbiome Consortium Annual Symposium, 2023, NC:* (Poster) **Yunan Hu**, Jason W. Arnold, Rebecca Zasloff, Anika Rueppell, Scott T. Magness, José M. Bruno-Bárcena, M. Andrea Azcarate-Peril. Galacto-oligosaccharides Enhance Intestinal Barrier Function.
7. *NORC Student Research Symposium, 2022, NC:* (Poster) **Yunan Hu**, Jason Arnold, Johanna Smeekens, Michael Kulis, Ellyce San Pedro, Scott Magness, Andrea Azcarate-Peril. Mechanism of Galectin-1 induction by Galactooligosaccharides.

4) Mentoring Activities

Undergraduate Student Supervised Research

The table below lists supervised undergraduates who gained research experience from me. The responsibilities of students include performing research on a defined project, demonstrating skills in standard scientific methods, evaluating scientific data, and scientific troubleshooting.

Student	Degree/activities	Dates	Currently at
Garrison Bruber	BS. Postbaccalaureate research	May 2026 -	UNC-Chapel Hill
Louis Lee	Biology	May 2026 -	UNC-Chapel Hill
Inhwan Choi	BBSP PhD rotation	Nov 2025 – Feb 2026	UNC-Chapel Hill
Angela Molina	Nutrition	Aug 2025-	UNC-Chapel Hill
Jin Ho Kim	Biology – Honor Thesis	May 2025-	UNC-Chapel Hill
Yoo Rim Lee	Nutrition - Honor Thesis	March 2025-May 2026	UNC-Chapel Hill
Weston Matthews	Biology	Aug. 2024- May 2025	UNC-Chapel Hill
Nehal Devpura	Nutrition - Honor Thesis	Feb. 2023- May 2026	University of Arizona medical school

Rebecca Zasloff	Biology - Honors Thesis	Aug. 2021 – Jun. 2022	Duke and Harvard school of public health
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5) Professional Service

2024-Present	President of UNC Microbiome Trainee club
2025-2027	Co-chair of Gordon Research Seminar (GRS): Lactic Acid Bacteria Biology, Symbioses and Applications
2025-Present	Leadership of Nutritional Microbiology ASN's Groups Engaging Members (GEMs)
2023-2024	UNC Biological & Biomedical Science Program (BBSP) Leadership Group
2025.8	Judge of iGEM Mini Jamboree in Healthcare category
2023.9-Present	Laboratory Animal Coordinator at UNC Microbiome Core
2023-Present	Reviewer, Gut Microbiome
2025-Present	Reviewer, Food Bioscience

Member since:

2022	International scientific association for probiotics and prebiotics Students and Fellows Association (ISAPP-SFA)
2024	American Society for Nutrition (ASN)
2024	North Carolina Branch of the American Society for Microbiology (NC-ASM)
2024	American Society for Microbiology (ASM)

Awards:

2022	Travel award for International scientific association for probiotics and prebiotics annual conference, Denver
2026	Travel award for International scientific association for probiotics and prebiotics annual conference, Japan